Math 651 Spring 2021

Take-home Final Exam

Due: Wednesday, April 28, 12:00 noon

In doing this exam, you may refer freely to your Math 651 class notes, the course textbooks, and any of the classes previous homework for this course, including any comments and notes on them. You may not use any other references, including web-based, printed, or human ones. The one exception to this rule is that you may refer to non-human TEX resources that might help you present your work better.

You are not to discuss your work with anyone other than your instructor. In particular you should not discuss the problems with any students, whether they are in this class or not, or any other professors. If you have any doubts as to what is allowed, you are responsible for asking for clarification.

Solutions should be presented with a good mathematical writing style, and must be typeset. Use a proposition/proof format whenever appropriate. If you need to use a result in the textbooks, be sure sure you explicitly refer to it, giving both its number and page number. Clarity of presentation will be a factor in grading.

The problems are not in order of difficulty, and you may not get them all.

1. (15 pts.) Let $\mathcal{S} = \{X_\alpha\}_{\alpha \in \mathcal{A}}$ be a collection of non-empty topological spaces, and let $P = \prod_{\alpha \in A} X_\alpha$.

   Theorem 1 (Munkres, p. 205)**.** *If P is Hausdorff, regular, or normal, then X_α has the same property.*

   Proof. Recall that projection $p : P \to X_\beta$ onto a particular factor $X_\beta \in \mathcal{S}$ is an open map.

   Suppose there are distinct points $x, y \in X_\beta$ and closed sets E, F such that $x \in E$ and $y \in F$. Then, there are distinct points $\vec{x}, \vec{y} \in P$ such that $p(\vec{x}) = x$ and $p(\vec{y}) = y$.

   Suppose P is Hausdorff. Then, there exist disjoint open neighborhoods U of $\vec{x}$ and V of $\vec{y}$ in P. Since p is a well-defined open map, it follows that $p(U)$ and $p(V)$ are disjoint open neighborhoods of x and y in X_β.

   Suppose P is regular, and suppose F does not contain x. Continuity of p implies that $p^{-1}(F)$ is closed set in P which does not contain $\vec{x}$. Since P is regular, there exist disjoint open sets U containing x and V containing $p^{-1}(F)$. Since p is a well-defined open map, we obtain disjoint open neighborhoods $p(U)$ of x and $p(V)$ of F.

   Suppose P is normal, and suppose E does not intersect F. Then, $p^{-1}(E)$ and $p^{-1}(F)$ are disjoint closed sets containing $\vec{x}$ and $\vec{y}$, respectively. Since P is normal, there are disjoint open sets U and V containing $p^{-1}(E)$ and $p^{-1}(F)$, respectively. Since p is a well-defined open map, we obtain disjoint open neighborhoods $p(U)$ of E and $p(V)$ of F.

   Note that many of the properties of P carry over to each factor X_β because of an underlying isomorphism between X_β and a subset of P that corresponds to a product of X_β with singleton subsets of X_α for $\alpha \neq \beta$. □

2. (15 pts.) Let $p : E \to B$ be a covering map.

Theorem 2. *The following are true:*

(a) p is open.

(b) If B is Hausdorff, then E is Hausdorff.

(c) E is compact if, and only if, B is compact and all fibers of p are finite.
(You may want to first assume B is connected. Establishing the claim under that assumption implies it more generally by applying it to each of the finitely-many connected components of B and their inverse images in E. But you'll need to explain why there are only finitely-many connected components of B.)

(d) If E is Hausdorff and compact, then B is Hausdorff.

Proof. Let $p : E \to B$ be a covering map.

(a) We show that an open set $U \subseteq E$ has an open image set $p(U) \subseteq B$ by showing that each point $b \in p(U)$ is contained in an open subset contained in $p(U)$.
Since p is a covering map, there is a neighborhood $V \subseteq B$ of b with disjoint open sets $\{V_\alpha\}_{\alpha\in A}$ equipped with homeomorphisms $p|_\alpha : V_\alpha \to V$ and satisfying $p^{-1}(V) = \bigsqcup_{\alpha\in A} V_\alpha$. By surjectivity of p, there is some $x \in U$ satisfying $p(x) = b$ and $x \in V_\beta \subseteq p^{-1}(V)$ for some $\beta \in A$. It follows that b is contained in the subset $p|_\alpha (V_\alpha \cap U) \subseteq p(U)$, which is an open since the homeomorphism $p|_\alpha$ is an open map, and $V_\alpha \cap U$ is open since U is open in E.
Repeating this procedure fr all $b \in p(U)$ shows that $p(U)$ can be written as a union of open sets. Therefore, $p(U)$ is open.

(b) We show that E is Hausdorff when B is Hausdorff.
If E is the singleton space, then it is automatically Hausdorff. Otherwise, let x, y be distinct points in E.
Suppose $p(x) = p(y) = b$ and let U be a neighborhood of b that is evenly covered via p by $p^{-1}(U) = \sqcup_{\alpha\in A} U_\alpha$ for disjoint neighborhoods U_α which are homeomorphic to U under p. In particular, x and y are contained in disjoint neighborhoods U_α.
Otherwise, suppose $p(x) \neq p(y)$. By the Hausdorff property, there are disjoint open sets U of $p(x)$ and V of $p(y)$. Since p is a covering map, there are open neighborhoods V_1 and V_2 of $p(x)$ and $p(y)$ which are evenly covered by p. They yield disjoint open neighborhoods $V_1 \cap U$ and $V_2 \cap V$ of $p(x)$ and $p(y)$ which are evenly covered by p so that $p^{-1}(V_1 \cap U) = \sqcup_{\alpha_i} E_{\alpha_i}$ and $p^{-1}(V_2 \cap V) = \sqcup_{\alpha_j} F_{\alpha_j}$. The E_{α_i} containing x does not intersect the F_{α_j} containing y. Thus, E is Hausdorff.

(c) Suppose B is connected.
Suppose instead that E is compact, and consider an open cover $\mathcal{B} = \{B_\alpha\}_{\alpha\in A}$ of B. By compactness of E, there exists a finite subcover $\mathcal{U} = \{p^{-1}(B_{\alpha_i})\}_{i=1}^n$ of the open cover $p^{-1}(B) = \{p^{-1}(B_\alpha)\}_{\alpha\in A}$ of E. Since p is an open surjection, the images $p(\mathcal{U}) = \{p(p^{-1}(B_i))\}_{i=1}^n = \{B_{\alpha_i}\}_{i=1}^n$ are a finite subcover of $\mathcal{B}$ in B. Each fiber has a finite subcover and discreteness implies that it is finite.
Now suppose instead that B is compact and all fibers of p are finite, and consider an open cover $\mathcal{U} = \{U_\alpha\}_{\alpha\in A}$ of E. For each b, there is a finite subcover $\{U_{b_i}\}_{i=1_b^n} \subseteq \mathcal{U}$ of the fiber $p^{-1}(b)$. Let V_b be an open neighborhood of b in B such that $p^{-1}(V_b) \subseteq \cup_{i=1}^{n_n} U_{b_i}$. Since $\{V_b\}_{b\in B}$ covers B, the compactness of B yields a finite subcover $\{V_{b_i}\}_{i=1}^m$ of B and its pre-image forms a cover of E, as well as $\{V_{b_k}\}_{i=1}^m$. This yields a finite subcover in $\mathcal{U}$ of E.
Notice that if E is compact, then B has finitely many connected as there is a finite subcover of an open cover containing a set with at least one element per connected component. If all fibers

of p are finite then B similarly has finitely many connected components. The result follows by induction on each connected component.

(d) Suppose E is Hausdorff and compact.

If B is a singleton space, then it is automatically Hausdorff. Otherwise, let $b, c \in B$ be distinct.

Since p is a well-defined surjection, there exist distinct points $x, y \in E$ such that $p(x) = b$ and $p(y) = c$.

By the Hausdorff axiom, there exist disjoint open neighborhoods U of x and V of y, and thus closed sets $X - U$ not containing x and $X - V$ not containing y.

Since p is closed map, $p(X - U)$ is a closed set not containing $p(x)$ and $p(X - V)$ is a closed set not containing $p(y)$. Thus, we obtain open neighborhoods $B - p(X - U)$ of b and $B - p(X - V)$ of c.

□

3. (10 pts.) There are analogs for homology of many results for the fundamental group.

Theorem 3. *Let A be a subset of X, and let $i : A \to X$ be the inclusion map. The following hold.*

(a) If $r : X \to A$ is a retraction, then the maps induced on all homology groups by the inclusion and retraction are injections and surjections, respectively.

(b) If A is a deformation retract of X, then the maps induced on all homology groups by the inclusion and deformation retraction are isomorphisms.

Proof. For some n, let $i_\star : H_n(A) \to H_n(X)$ be the map induced by inclusion and let $r_\star : H_n(X) \to H_n(A)$ be the map induced by a retraction $r : X \to A$.

(a) Let $z \in \ker(i_\star)$. Then, $(r_\star \circ i_\star) = r_\star(i_\star(z)) = i_\star(x) = 0$, so $z \in \ker(r_\star \circ i_\star)$ implies the following containment of groups $\ker(i_\star) \subseteq \ker(r_\star \circ i_\star) = 1_{H_n(A)} = \{0\}$. Since $\ker(i_\star) = 0$, the map $i_\star$ is injective.

Let z be in $H_n(A)$. Then, it is a cycle in $Z_n(A) \subseteq Z_n(X)$ with a boundary in $B_n(A) \subseteq B_n(X)$. Therefore, z is in $H_n(X)$. Thus, $r_\star$ is surjective.

(b) Suppose there is a deformation retraction $H : X \times I \to X$ of X onto A, i.e. a homotopy from $1_X = H(x, 0) : X \to X$ to $r = H(x, 1) : X \to X$. It suffices to show that $i_\star$ is surjective and that r is injective.

According to Hatcher Theorem 2.10, the homotopy-equivalent maps $[i \circ r] = [1]$ induce the same homomorphism $(i \circ r)_\star = 1_{X_\star} : H_n(X) \to H_n(X)$. Therefore, $i_\star$ is surjective and $r_\star$ is surjective.

□

4. (30 pts.) In this problem, we will take the first steps to understanding the higher homotopy groups, $\pi_n(X, x_0)$, for $n \geq 2$. Here $x_0 \in X$ is a fixed base point.

Let $I^n = I \times \cdots \times I$ be the n-fold product of intervals $I = [0, 1]$, and let ∂I^n denote its boundary in $\mathbb{R}^n$. We say f is a map of $(I^n, \partial I^n)$ into (X, x_0), and write

$$f : (I^n, \partial I^n) \to (X, x_0),$$

if $f : I^n \to X$ and $f(\partial I^n) = \{x_0\}$. We say two continuous maps $f, g : (I^n, \partial I^n) \to (X, x_0)$ are homotopic if there exists a homotopy

$$H : I^n \times I \to X$$

of f to g for which $H(x, t) = x_0$ if $x \in \partial I^n$. (More formally, we should say the maps are *homotopic relative to* ∂I^n to indicate the homotopy fixes ∂I^n, but for simplicity we drop the extra phrase here.)

It's straightforward to show that this notion of homotopic defines an equivalence relation, so let $[f]$ denote the equivalence class of such an f.

Theorem 4. *Let $\pi_n(X, x_0)$ be the collection of homotopy equivalence classes, which is, so far, only a set, and not yet a group. The following statements (a–e) are true.*

(a) If X is a convex subset of $\mathbb{R}^n$, then $\pi_n(X, x_0)$ has a single element.

(b) Define an operation $*$ on continuous maps of $(I^n, \partial I^n)$ into (X, x_0), by

$$f * g(t_1, t_2, \ldots, t_n) = \begin{cases} f(t_1, t_2, \ldots, t_{n-1}, 2t_n) & \text{if } t_n \leq 1/2, \\ g(t_1, t_2, \ldots, t_{n-1}, 2t_n - 1) & \text{if } t_n \geq 1/2. \end{cases}$$

The product $f * g$ is a continuous map of $(I^n, \partial I^n)$ into (X, x_0), and $*$ induces a well-defined operation on homotopy classes by $[f] * [g] = [f * g]$.

(c) The class $[e_{x_0}]$ of the constant map to x_0 is a left identity element for $*$ on homotopy classes.

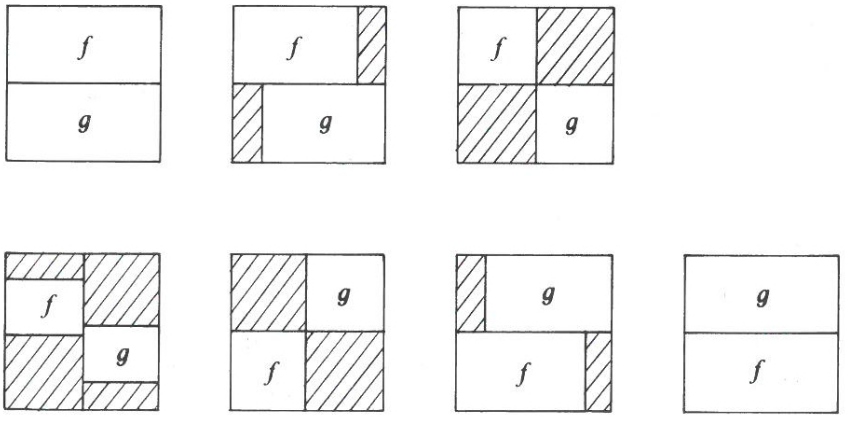
(d) For a continuous map $f : (I^n, \partial I^n) \to (X, x_0)$, continuous function $g : (I^n, \partial I^n) \to (X, x_0)$ given by the formula

$$g(t_1, t_2, \ldots, t_n) = f(t_1, t_2, \ldots, t_{n-1}, 1 - t_n)$$

satisfies $[f] * [g] = [e_{x_0}] = [g] * [f]$. Give a proof that $[f] * [g] = [e_{x_0}]$. (We will not prove $[g] * [f] = [e_{x_0}]$.)

Assuming $*$ is associative on homotopy classes (which I will NOT ask you to prove), the last 3 subproblems show $\pi_n(X, x_0)$ is a group.

(e) For $n > 1$, $\pi_n(X, x_0)$ is always abelian. The following pictures hint at how this can be proved for the $n = 2$ case. To show you understand, give a formula for a homotopy that does what is depicted in transitioning across the top 3 pictures *only.*



Proof. Let X be a space.

(a) Suppose X is a convex subset of $\mathbb{R}^n$. For any two continuous maps

$$f, g : (I^n, \partial I^n) \to (X, x_0),$$

we construct a homotopy $H : I^n \times I \to X$ from f to g for which $H(x,t) = x_0$ if $x \in \partial I^n$.
In particular, consider the straight-line map given by $H(s,t) = (1-t)f(s) + tg(s)$ for $s \in I^n$ and $t \in I$. Convexity of X implies that H is a homotopy. If $x \in \partial I^n$, then we obtain:

$$H(x,t) = (1-t)f(x) + tg(x) = (1-t)x_0 + tx_0 = (1-t+t)x_0 = x_0.$$

Since any two continuous maps are homotopic, it therefore follows that the homotopy group $\pi_n(X, x_0)$ has a single element.

(b) We show the product $f * g$ is a continuous map of $(I^n, \partial I^n)$ into (X, x_0) by applying the closed pasting lemma, and then show the operation $*$ is well defined on homotopy classes by constructing the required homotopy.
Notice that the product is a continuous function defined on two closed sets: $A = I^{n-1} \times \left[0, \frac{1}{2}\right]$ and $B = I^{n-1} \times \left[\frac{1}{2}, 1\right]$. On each closed set, its continuity follows from the fact that it is equal to a composition of continuous functions, so the operation $*$ is continuous on I^n by the closed pasting lemma. If $x \in \partial I^n$, then $(f * g)(x) \in \{f(x), g(x)\} = \{x_0\}$, so $f * g$ is a continuous map of $(I^n, \partial I^n)$ into (X, x_0).
Next we show that $[f] = [h]$ and $[g] = [k]$ together imply that $[h] * [k] = [h * k] = [f * g] = [f] * [g]$. In particular, suppose there are homotopies H and K from f to h, and from g to k. Then, there is a homotopy $(H * K) : I^n \times I \to X$

$$(H * K)(t_1, \ldots, t_n, t) = \begin{cases} H(t_1, t_2, \ldots, t_{n-1}, 2t_n, t) & \text{if } t_n \leq 1/2, \\ K(t_1, t_2, \ldots, t_{n-1}, 2t_n - 1, t) & \text{if } t_n \geq 1/2. \end{cases}$$

which achieves the following values:

$$\begin{aligned} (H * K)(\cdot, 0, t) &= H(\cdot, 0, t) = f(\cdot, 0) = (f * g)(\cdot, 0) \\ (H * K)(\cdot, 1/2, t) &= H(\cdot, 1/2, t) = g(\cdot, 1) = h(\cdot, 0) = K(\cdot, 1/2, t) \\ (H * K)(\cdot, 1, t) &= K(\cdot, 1, t) = k(\cdot, 1) = (h * k)(\cdot, 1). \end{aligned}$$

A block diagram would depict this as H on the left half with f on the lower quarter and h on the upper quarter, and K on the right half with g on the lower quarter and k on the upper quarter. Reading this from the bottom up expresses that this is a homotopy from $f * g$ to $h * k$. Alternatively, reading this from left to right expresses something like a homotopy from f and h to g and k, although this would occur in a coordinate not supported by our definition of a homotopy which only encapsulates the final coordinate.

(c) Then,

$$\begin{aligned} [e_{x_0} * f](t_1, t_2, \ldots, t_n) &= \begin{cases} [e_{x_0}](t_1, t_2, \ldots, t_{n-1}, 2t_n) & \text{if } t_n \leq 1/2, \\ [f](t_1, t_2, \ldots, t_{n-1}, 2t_n - 1) & \text{if } t_n \geq 1/2. \end{cases} \\ &= \begin{cases} x_0 & \text{if } t_n \leq 1/2, \\ [f](t_1, t_2, \ldots, t_{n-1}, 2t_n - 1) & \text{if } t_n \geq 1/2. \end{cases} \\ &= [f]. \end{aligned}$$

(d) Notice that

$$[f * g](t_1, t_2, \ldots, t_n) = \begin{cases} [f](t_1, t_2, \ldots, t_{n-1}, 2t_n) & \text{if } t_n \leq 1/2, \\ [g](t_1, t_2, \ldots, t_{n-1}, 2t_n - 1) & \text{if } t_n \geq 1/2. \end{cases}$$
$$= \begin{cases} [f](t_1, t_2, \ldots, t_{n-1}, 2t_n) & \text{if } t_n \leq 1/2, \\ [f](t_1, t_2, \ldots, t_{n-1}, 1 - (2t_n - 1))) & \text{if } t_n \geq 1/2. \end{cases}.$$

The first $(n-1)$ coordinates are contractible, and the nth forms a contractible loop as well.

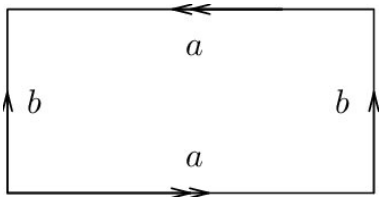
(e) We construct a homotopy $H : I^2 \times I \to X$ where $H(t_1, t_2, 0)$ denotes the first square and $H(t_1, t_2, 1)$ denotes the third square, and t_3 moves through the page. Let t_1 be the coordinate which traverses a given cross-section from left to right and t_2 be the coordinate which traverses a given cross-section from bottom to top.

Then, the homotopy is given below.

$$\begin{cases} \begin{cases} x_0 & \text{for } t_1 \leq t_3 \\ g((1 - e^(t_1 - t_3))/(t_1 - e^(1 - t_3)), 2t_2) & \text{for } t_3 \leq t_1 \leq 1 \end{cases} & \text{for } t_2 \leq 1/2 \\ \begin{cases} f(t_1/t_3, 2t_2 - 1) & \text{for } t_1 \leq t_3 \\ x_0 & \text{for } t_1 \geq t_3 \end{cases} & \text{for } t_2 \geq 1/2 \end{cases}$$

□

5. (30 pts.) Let X be the *Klein bottle*, a 2-dimensional surface obtained as a quotient of a rectangle $R = A \times B$ by identifying edges as shown in the figure below. Let a and b be loops in X based at x_0, with images as shown in the figure, so $[a], [b] \in \pi_1(X, x_0)$.



Theorem 5. *The following are true:*

(a) The space X has $\mathbb{R}^2$ as its universal cover

(b) The space X has T as a 2-fold cover, and this can be seen through the universal covers of X and T.

(c) In $\pi_1(X, x_0)$ neither $[a]$ nor $[b]$ is the identity, nor equal to their own inverses, and that $[a] * [b] = [b] * [a]^{-1}$.

(d) The elements of $\pi_1(X, x_0)$ can be identified with elements of $\mathbb{Z}^2$ as a set*, but the correspondence does* not *yield a group isomomorphism.*

(e) Using this identification with $\mathbb{Z}^2$, the group operation in $\pi_1(X, x_0)$ is expressed as

$$(n, m)(k, \ell) = (n + \phi_m(k), m + \ell),$$

where $\phi_m : \mathbb{Z} \to \mathbb{Z}$ is the identity map for m even, and the inverse map for m odd.

(f) For any two groups G,H, with $\mathrm{Aut}(G)$ *denoting the group of automorphisms of G, let*

$$\phi : H \to \mathrm{Aut}(G)$$

be any homomorphism. Then the semidirect product $G \rtimes_\phi H$ with respect to ϕ *is defined as the set $G \times H$ together with the operation*

$$(g_1, h_1)(g_2, h_2) = (g_1 \phi(h_1)(g_2), h_1 h_2).$$

The semidirect product $\mathbb{Z} \rtimes_\phi \mathbb{Z}$ is defined for two ϕ maps, and one of them is the fundamental group of X.

Proof. Let X be the Klein bottle and recall that the plane $\mathbb{R}^2$ is simply connected.

(a) We explain informally why X has $\mathbb{R}^2$ as its universal cover by showing how to tile the plane with the rectangle R as in Figure 1.
Without loss of generality, we assume $x_0 = (0, 0)$ and that the rectangle is given by $R = A \times B$, where $A = [0, h]$ and $B = [0, v]$. Let $e_0 = (0, 0)$ and $p(e_0) = x_0$. Let $p(e) = q(e)$, where q is the quotient map of the rectangle R and e is in R. We tile the band $B_1 = \mathbb{R} \times B \subseteq \mathbb{R}^2$ by repeating copies of R, shifted horizontally by $|A|$. We tile the band $B_2 = \mathbb{R} \times [v, 2v]$ by reflecting each rectangle in B_1 horizontally and then translating them up. Since each point e in the plane is in some rectangle, each point has a representative $[e]$ in R, so we assign $p(e) = p([e])$.
The basepoint x_0, for example, is covered by $(0, 0)$ as well as (e_1, e_2) for multiples $e_1 = nh$ and $e_2 = mh$ for all integers $n, m \in \mathbb{Z}$. The point in the center of the rectangle $(h/2, v/2)$ is covered by points of the form $(nh + h/2, mh + v/2)$ for all integers $n, m \in \mathbb{Z}$. Any point in the Klein bottle has an associated point on the rectangle, and its fiber is given by the coset of the fiber of the basepoint added to the corresponding point on the rectangle.
This covering is notably different from the torus which tiles the plane via translations of isomorphic rectangles; we instead tile it with translations and reflections.

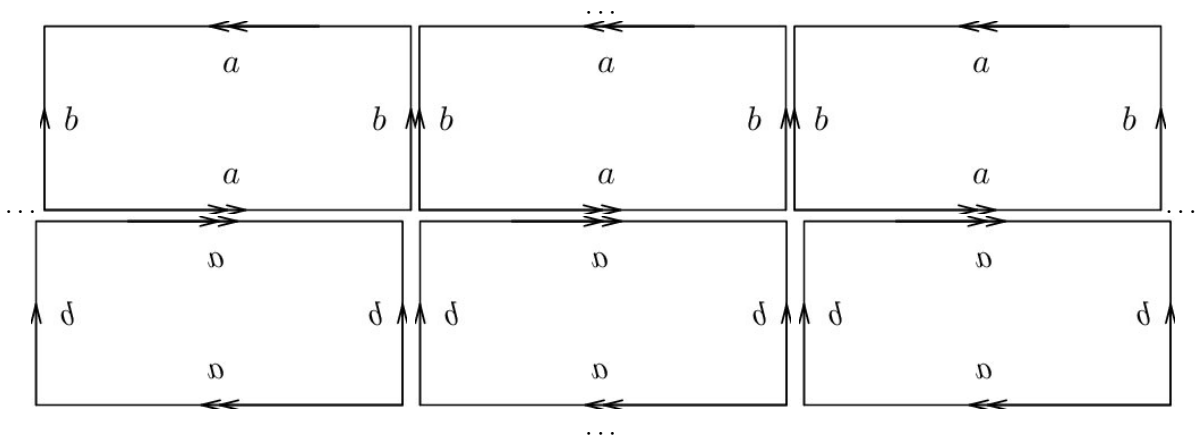


Figure 1: Caption

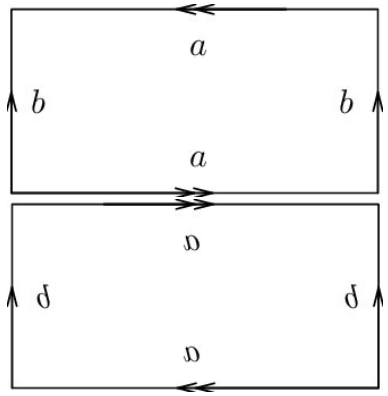


Figure 2: Caption

(b) Consider a horizontal reflection of the rectangle stacked on top of the rectangle, as in Figure 2, with the overlapping edges a identified – the quotient map would then necessarily identity the four outer corners as one, the two vertical edges as one, and the two horizontal edges as one. This shows how to see the torus as a copy of two klein bottles. Alternatively, the covering space of the torus can be seen as the covering space of the klein bottle with vertical units multiplied by $2v$ instead of v.

(c) Following the diagram, it is clear that $[a] * [b] * [a]^{-1} = [b]$ as there is a homotopy which slowly contracts the rectangular loop $[a] * [b] * [a]^{-1} * [b]^{-1}$ to the constant loop $[e_{x_0}]$.

By the Seifert-van Kampen theorem, the fundamental group of X is isomorphic to $\mathbb{Z}^2/\langle abab^{-1}\rangle$, which notably lacks relations which restrict the order of powers of a or powers of b. This can be visualized even without the van Kampen theorem.

Therefore, they each have infinite order. If either were equal to the identity or their own inverses, they would have order 1 or 2, which would be contradictions.

(d) A loop in X can be lifted to a path in the universal cover which starts at $(0, 0)$ and ends at (nu, mv) for unique integers $n, m \in \mathbb{Z}$, and thus elements of the fundamental group can be identified with elements (n, m) of the set $\mathbb{Z}^2$. However, $\mathbb{Z}^2$ is an abelian group, while the result in part (c) show

that $[a]$ and $[b]$ fail to commute:

$$[a] * [b] = [b] * [a]^{-1} \neq [b] * [a].$$

Since commutativity is a property preserved by group isomorphisms, it follows that the set correspondence above is not a group isomorphism.

(e) The multiplication $([a]^n * [b]^m)([a]^k * [b]^l)$ corresponds to starting at $[a]^n * [b]^m$, moving l units in the direction of $[b]$ which is positively upward and moving k units in the direction of $[a]$ which is positively right-ward if m is even and positively left-ward (negatively right-ward) if m is odd.

This is precisely captured by the formula $(n, m)(k, l) = (n + \phi_m(k), m + l)$, where $\phi_m : \mathbb{Z} \to \mathbb{Z}$ is the identity map for even m and the inverse map for odd m.

(f) We show (i) that there are two maps $\phi_{\pm 1} : \mathbb{Z} \to \mathrm{Aut}(\mathbb{Z})$ with respect to which the semidirect product $\mathbb{Z} \rtimes_\phi \mathbb{Z}$ is defined, and (ii) conclude that each of them yield the fundamental group of a known space.

i. First, we show there are two automorphisms $\psi_{\pm 1} : \mathbb{Z} \to \mathbb{Z}$ in $\mathrm{Aut}(\mathbb{Z})$ by recalling that an arbitrary automorphism $\psi : Z \to \mathbb{Z}$ is uniquely determined by the value assigned to a generator, say $\psi(1)$ without loss of generality. Surjectivity demands $\psi(\mathbb{Z}) = \psi(1)\mathbb{Z} = \mathbb{Z}$ and thus that $\psi_{\pm 1}(1) = \pm 1$ which extend to $\psi_{\pm 1}(n) = \pm n$.
Next we show there are two homomorphisms $\phi_{\pm 1} : \mathbb{Z} \to \mathrm{Aut}(\mathbb{Z})$, again by recalling that an arbitrary homomorphism $\phi : \mathbb{Z} \to \mathrm{Aut}(\mathbb{Z})$ is uniquely determined by the value assigned to a generator, say $\phi(1)$ without loss of generality. Without loss of generality, we obtain $\phi_{\pm 1}(1) = \psi_{\pm 1}$. Clearly, $\phi_1(h_1)(g_2) = \psi_1(g_2) = g_2$. By linearity, we obtain $\phi_{-1}(h_1)(g_2) = \psi_{(-1)^{h_1}}(g_2) = (-1)^{h_1} g_2$.
The semidirect product $\mathbb{Z} \rtimes_{\phi_1} \mathbb{Z}$ with respect to ϕ_1 is equipped with the following operation

$$(g_1, h_1)(g_2, h_2) = (g_1 + \phi_1(h_1)(g_2), h_1 + h_2) = (g_1 + \psi_1(g_2), h_1 + h_2) = (g_1 + g_2, h_1 + h_2),$$

which is precisely the operation on the fundamental group $\mathbb{Z} \times \mathbb{Z}$ of the torus.

ii. Let $\phi = \phi_{-1}$. Then, the semidirect product $\mathbb{Z} \rtimes_\phi \mathbb{Z}$ with respect to ϕ is equipped with the following operation

$$(g_1, h_1)(g_2, h_2) = (g_1 + \phi(h_1)(g_2), h_1 + h_2) = (g_1 + \phi_{h_1}(g_2), h_1 + h_2),$$

where $\phi_{h_1}(g_2) = \phi(h_1)(g_2) = \psi_{(-1)^{h_1}}(g_2) = (-1)^{h_1} g_2$ is the identity map for even h_1 and the inverse map for odd h_1.

Up to relabeling, the semidirect product $\mathbb{Z} \rtimes_{\phi_{-1}} \mathbb{Z}$ with respect to ϕ_{-1} is precisely the group operation in $\pi_1(X, x_0)$.

Therefore, the fundamental group of the Klein bottle is the uniquely non-trivial semidirect product of $\mathbb{Z}$ with itself. □